

Joseph M. Macedonia

Department of Biological
Anthropology, Duke University,
Durham, N.C., USA

The Vocal Repertoire of the Ringtailed Lemur (*Lemur catta*)

Key Words

Vocal repertoire
Ringtailed lemur
Lemur catta
Communication
Calls
Vocal development

Abstract

The vocal repertoire of the ringtailed lemur (*Lemur catta*) is described quantitatively and spectrographically. Structure, usage and inferred function of the vocalizations of adults and immatures are examined and discussed. The size, acoustic variability and development of the ringtailed lemur's vocal repertoire are compared to prior assessments for this species as well as for other primates. With up to 22 different adult call types, the ringtailed lemur has a moderately large vocal repertoire for a primate.

Introduction

Understanding vocal communication in any species requires documenting its vocal repertoire. Only by describing the range of sounds produced, and how they relate to one another in structure, usage and function, can the extent of a species' ability to communicate vocally be assessed. To date, the vocal repertoires of many primate species have been published [1–14; see 15 for references prior to 1984], although these studies represent only a fraction of the extant primate taxa. Among these species, the Malagasy lemurs are not well represented.

Within the past three decades, Andrew [16] described some vocalizations of several lemur species, while Petter and Charles-Dominique [17] provided an overview of vocal communication in lemurs and other prosimians. In a comparative behavioral study of ringtailed lemurs (*Lemur catta*) and white sifakas (*Propithecus verreauxi verreauxi*), Jolly [18] gave verbal descriptions of their vocal repertoires but referred the reader to Andrew [16] for available spectrographic details. In addition to vocalizations produced in the context of predators [19–23], information on other aspects of vocal communication in lemurs has become increasingly available during the

past decade [24–31]. Nevertheless, vocal repertoires for only three lemur species have been published so far (*Eulemur fulvus albifrons* [32], *Varecia variegata variegata* [10] and *Daubentonia madagascariensis* [14]), and only the latter study is quantitative. Moreover, research on vocal development in the lemuriform primates has been virtually absent [but see 33]. This report on vocal communication in the ringtailed lemur is a step toward a greater representation of the lemurs among those primates whose vocal repertoires have been described.

Methods

Research Environment

Observations and recordings were made within Natural Habitat Enclosure No. 2 (NHE-2: 3.3 ha) at the Duke University Primate Center (DUPC, Durham, N.C., USA). NHE-2 is an area of mixed pine and hardwood forest surrounded by a mildly electrified fence. A 10-m cleared strip between the forested enclosure and the fence prevents lemurs from escaping. The lemurs maintained in this enclosure received a daily supply of monkey chow and were provisioned twice weekly with mixed cut fruit, although they spent much of their time foraging on the local plants [34].

Raptors (e.g. red-tailed hawk, *Buteo jamaicensis*; red-shouldered hawk, *Buteo lineatus*; great-horned owl, *Bubo virginianus*), carnivores (gray fox, *Urocyon cinereoargenteus*; raccoon, *Procyon lotor*; weasel, *Mustela* sp.), venomous snakes (copperhead, *Agkistrodon contortrix*; cottonmouth, *Agkistrodon piscivorus*) and a nonvenomous, semiarboreal constrictor that could be dangerous to infants (black rat snake, *Elaphe obsoleta*) inhabited or frequented the forested enclosures. By June 1992, 10 lemurs had been killed by these predators: 1 young adult female ringtailed lemur by a gray fox, 5 infant ringtailed lemurs by a great-horned owl, 1 infant and 1 young adult male ringtailed lemur by a copperhead and 2 ruffed lemur newborns by a weasel. Whereas no lemur at the DUPC had been killed by a diurnal raptor, red-tailed hawks do attack mammals in the natural habitat enclosures [pers. observation] and are capable of taking prey up to the size and weight of adult *L. catta* (e.g. jackrabbits, *Lepus californicus* [pers. observation]).

Study Animals

The ringtailed lemur study group (Lc1 group) was introduced into NHE-2 in 1983 after living in another enclosure (NHE-1: 0.5 ha) since 1981. All members of Lc1 group other than the oldest male (RO) and female (LY) had been born in the enclosures or had lived there since weaning. During the study period (1986–1989), Lc1 group contained 21–33 individuals. Another group of ringtailed lemurs (Lc2 group) inhabited an adjacent enclosure (NHE-4: 5.8 ha) with a common boundary. Lc2 group was observed and recorded opportunistically (e.g. when interacting with Lc1 group at the intermediary fence) and during other studies [35]. Lc1 group shared enclosure NHE-2 with a group of black-and-white ruffed lemurs (*V. variegata variegata*) and a group of brown lemurs (*E. fulvus fulvus*) during 1986. A group of red-fronted lemurs (*E. fulvus rufus*) replaced the brown lemur group in 1987. Other than aggregating at feeding stations, the three species ranged independently. Subjects were fitted with color-coded nylon collars (denoting their birth year) with individually distinctive pendant tags. Individuals less than 1 year of age were classed as infants and those between 1 and 2 years as juveniles. Although adolescents (2–3 years of age) are also distinguished in the spectrograms, individuals over 2 years of age were classified as adults for quantitative analyses.

Recording Procedure

Vocalizations were recorded with a Sony TC-D5M cassette tape recorder using a Sennheiser ME-88 directional microphone. Except for calls possessing transient high frequencies (16–21 kHz), all *L. catta* vocalizations fell within the linear response range of this equipment (50 Hz to 15 kHz).

Recording/observation sessions typically lasted 2 h per recording day and were distributed across all daylight hours. Calls were recorded by standing 2–4 m from a subject (or a small cluster of individuals). Information dictated onto the tape following a vocal event included the caller's identity and details about the calling context. Approximately 85 h were spent on 46 days recording Lc1 group between April 1986 and June 1989. The majority of recordings were made during two periods: summer/fall 1987 (16 days) and spring/summer 1989 (25 days). Vocal development was the focus of the latter sampling period, with infants being recorded at a median interval of every 5 days (range = 1–12 days; $n = 24$) from March to June 1989. An additional 45 h were spent photographing behavior that

Table 1. Format for *L. catta* vocal repertoire: vocalization descriptions

Vocalization class	Calls fall into 3 broad classes for adults (affiliative vocalizations, agonistic vocalizations, alerting and antipredator vocalizations) and 2 classes for infants (affiliative vocalizations, distress vocalizations)
Vocalization	The name assigned to a call type; names assigned by previous authors that are maintained here give the earliest author's name and publication year
Synonyms	Call names of previous authors not used here are acknowledged as synonyms
Vocalizers	Sex and age class of call producers
Spectrograms	Figures in which calls are spectrographically illustrated
General structure	Three types of general acoustic structure are distinguished
Simple tonal structure	A call containing one or more clear harmonic bands
Simple atonal structure	A call lacking clear harmonic bands
Complex structure	Synchronous or asynchronous mixture of tonal and atonal sound energy
Frequency modulation	Patterns of frequency modulation, if present
Harmonics	Typical range of harmonics bands observed at a recording distance of 2–4 m
Additional characteristics	Noteworthy additional structural attributes
Contexts of emission	Context(s) in which a vocalization is emitted
Examples of usage	Specific examples of call usage
Inferred function	Apparent function of a vocalization

accompanied vocalizations. The time period formally devoted to documenting the vocal repertoire of *L. catta* underestimates the total time spent observing and recording this species, much of which occurred during prior [25] and simultaneous [19, 21, 23, 35] studies.

Call types were initially distinguished by ear, and the contexts in which they occurred were noted. Repeated observations of call/context associations, plus review of audio tapes and inspection of the spectrograms from all calls recorded, gave rise to the call classification scheme presented herein. The vocalizations are distinguished one from another by acoustic structure, and they are grouped into categories according to contexts of emission (see below). Where possible, structural and contextual aspects of the vocalizations were cross-referenced with previously published work [16–18].

Sound Analysis

All calls recorded were examined using a Uniscan II digital real-time audiospectrograph, but only calls of reasonably good recording quality were measured. This sometimes resulted in small samples for call types that were emitted in agonistic contexts (due to the rustle of leaf litter and vocalizations from several rapidly moving individuals), calls emitted by individuals primarily when arboreal and rarely emitted calls. The Uniscan II uses a fast Fourier transform of 256 points for all analysis scales. Call measurements were made at the narrowest frequency range possible (given the variable) to maximize measurements accuracy. The degree of measurement error ($\pm$ Hz) associated with each analysis scale was as follows: 0–2 kHz, $\pm$ 14.6 Hz; 0–5 kHz, $\pm$ 36.6 Hz; 0–10 kHz, $\pm$ 73.2 Hz; 0–20 kHz,

± 146.3 Hz. All spectrograms illustrated are 256-point transforms and were acquired with the 8-bit digitizing package 'MacRecorder' (Farallon, Inc.). The format used to describe adult and immature vocalizations is presented in table 1.

Descriptive Statistics

Descriptive statistics of call measurements were obtained using SPSS-PC. Sample size, median and range are presented in the Appendix for all call variables measured. Additional statistics are provided where relevant. Actual variables measured depend on call structure (tonal/atonal, pulsed/not pulsed) and include: (a) number of calls analyzed, (b) number of individuals sampled, (c) number of calls sampled per individual, (d) call duration in seconds, (e) start, end, high and low frequency in Hz for the fundamental frequency band, (f) number of units measured per call (if multi-unit), (g) unit duration, (h) interunit interval, (i) lowest and highest frequency of the major energy (darkest continuous portion of the spectrogram) in Hz. 'Calls' in quotes indicates that the vocalization contains a variable number of relatively uniform sound units. Such 'calls' are distinguished from vocalizations that consist primarily or solely of a single unit.

To obtain a rough estimate of call amplitude, *L. catta* vocalizations were broadcast over a playback speaker (Acoustic Research 'Powered Partners'). Amplitude was measured at 1 m with a realistic sound level meter ('A' weighting, fast response) and readings were taken as playback amplitude was 'titrated' to achieve the most natural-sounding level. From the range of decibel readings obtained, 9 amplitude categories were defined (table 2). The amplitude index for each call type is presented in the Appendix.

Results

Adult Vocalizations

Affiliative Vocalizations

Vocalization: 'Moan' [16]. Synonyms: 'mew' [18], 'cohesion miaouw' [17], 'cohesion call' [25]. Vocalizers: all except infants less than about 14 weeks of age. Spectrograms: figure 1a, b (see also [16], his fig. 11a; and [17], their fig. 11b).

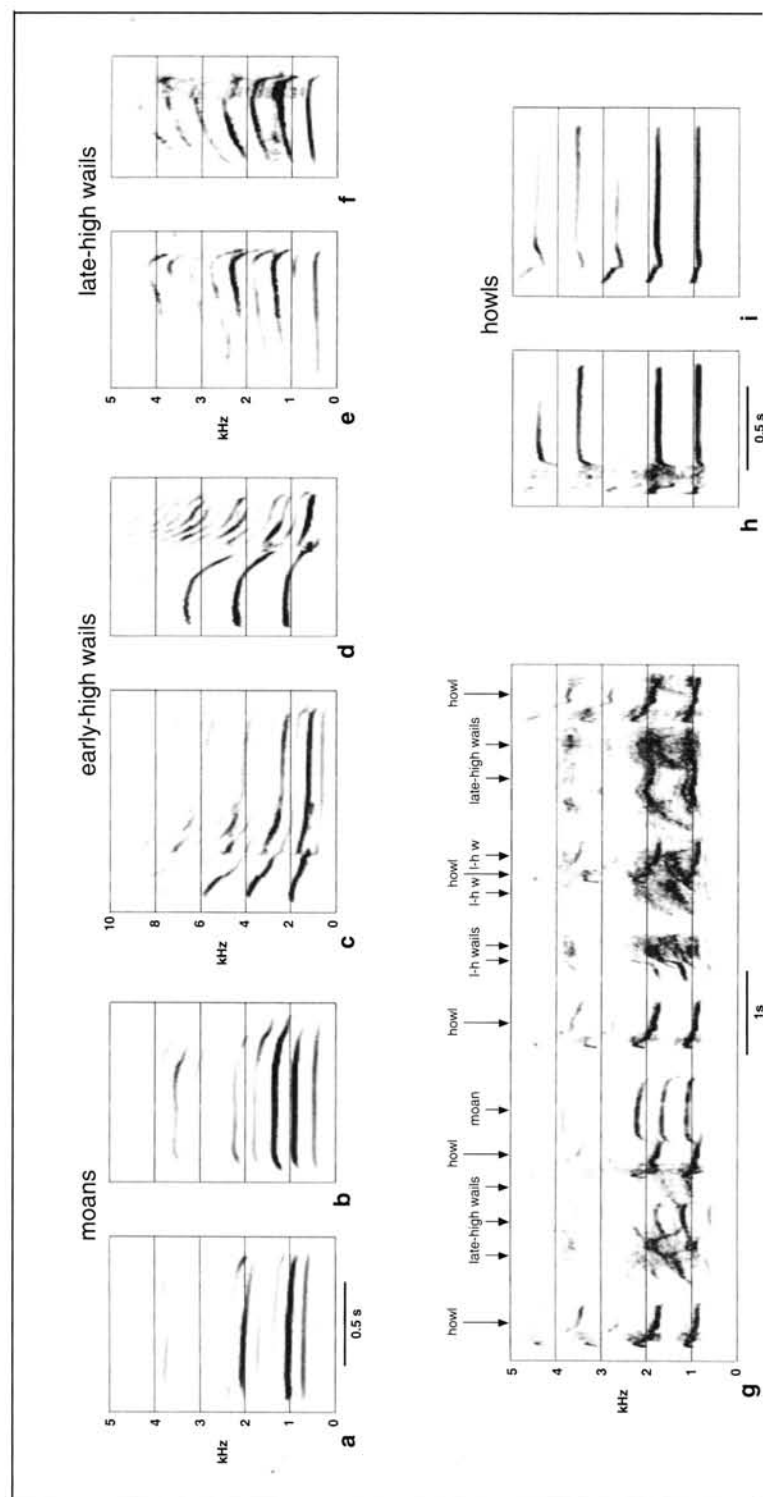
Table 2. Amplitude index (see Appendix)

Category	dB range at 1 m
(1) Extremely low	55–59
(2) Very low	60–64
(3) Low	65–69
(4) Low/moderate	70–74
(5) Moderate	75–79
(6) Moderate/high	80–84
(7) High	85–89
(8) Very high	90–94
(9) Extremely high	95–100

General structure: simple tonal; single unit. Frequency modulation: slight upswEEP is typically present at call onset, and equivalent downswEEP is seen at the offset. Harmonics: 2–4 strong bands with several weaker, higher bands. Additional characteristics: the moan is structurally intermediate between the two wail variants (see below).

Contexts of emission: emitted antiphonally as a contact call under conditions of low-to-moderate arousal. Examples of usage: (a) given as a relaxed cascade of vocal exchanges in a calm and relatively stationary group; (b) emitted commonly as a vocal response to other contact calls; (c) moans were always exchanged between Lc1 and Lc2 groups when meeting at the intermediary fence separating them. Inferred function: Promotes group cohesion in low-to-moderate arousal contexts.

Vocalization: 'Wail' [16], variant 1 (Early-High). Synonyms: 'me-ow' [18]. Vocalizers: all except infants less than about 6–8 weeks of age. Spectrograms: figure 1c, d (see also [16], his fig. 11b).



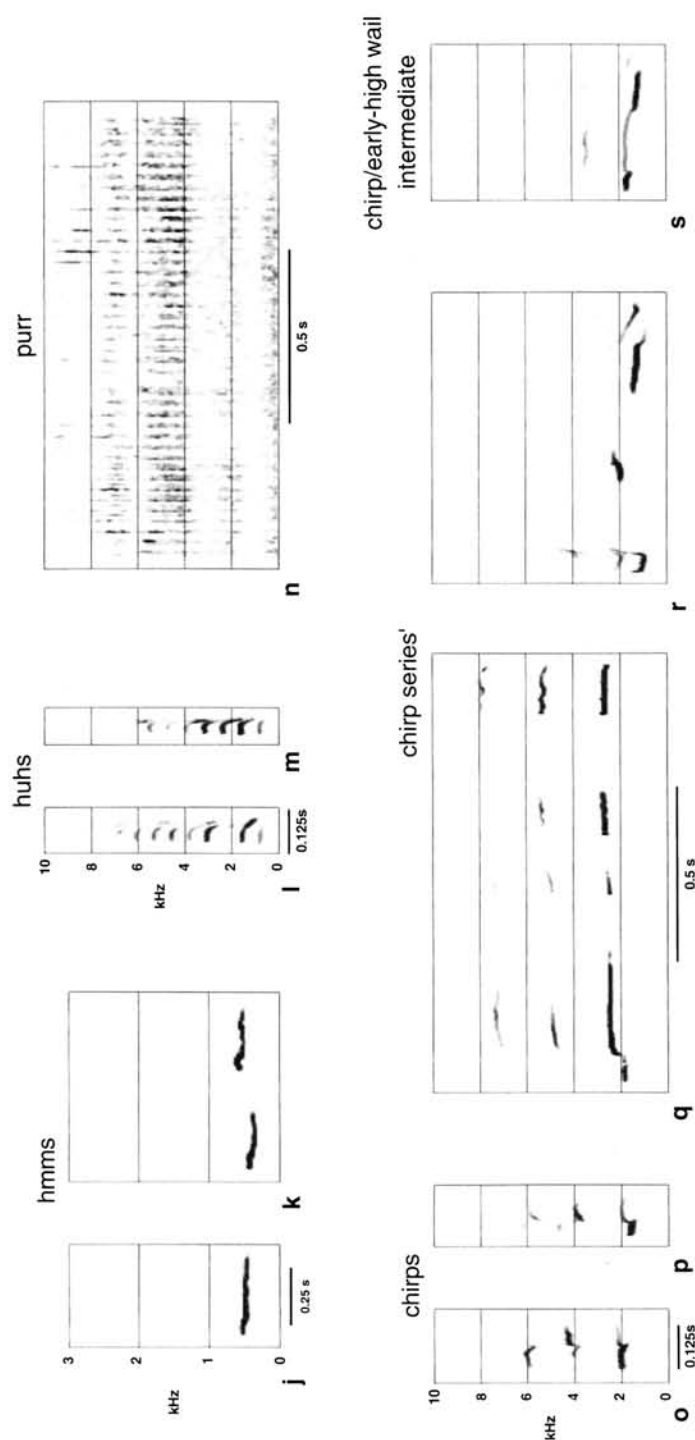


Fig. 1. Affiliative vocalizations. Spectrograms without frequency and/or time scales have the same scales as the preceding spectrogram for which this information is shown.

General structure: simple tonal; single unit. Frequency modulation: a single frequency downsweep of variable slope; in extreme examples the FM downslope exhibits one or two sudden temporal shifts in energy. Harmonics: 2–4 strong bands with several weaker, higher bands. Additional characteristics: can intergrade with chirps (fig. 1s).

Contexts of emission: contact in contexts of moderate arousal or excitement. Examples of usage: (a) emitted just prior to and during initial stages of daily provisioning; (b) given when sighting an approaching group member that had been visually isolated from the main group for a period of time. Inferred function: promotes group cohesion; indicates moderate-to-high arousal level of caller.

Vocalization: 'Wail' [16], Variant 2 (Late-High). Synonyms: 'me-ow' [18]. Spectrograms: figure 1e–g. Vocalizers: noninfant females (typically), males (rarely).

General structure: simple tonal to complex; single unit. Frequency modulation: slight-to-moderate frequency upsweep, falling more sharply at end of call. Harmonics: typically 2–3 strong bands with 3–4 weaker, higher bands. Additional characteristics: usually given in a 2-call sequence where a late-high wail is followed by an early-high wail, although sequences can contain up to 4 calls with one or more late-high wails followed by one or more early-high wails. In response to increasingly vigorous howling by males (see below), adult female late-high wails begin to exhibit harmonic 'side-bands' (caused by rapid frequency and/or amplitude modulation [36]) that eventually give way to total distortion due to calling with maximum effort (fig. 1g).

Contexts of emission: highest-arousal-level contact call; emitted less often than wail variant 1. Examples of usage: (a) emitted habitually by adult females in response to howls and frequently in response to yaps; (b) given occasionally by individuals in seeming attempts to contact 'missing' group-mates; (c) emitted by 'lost', individuals [M. Pereira, pers. commun.]. Inferred function: unclear, but may

promote group cohesion under conditions of extreme arousal.

Vocalization: 'Howl' [16]. Vocalizers: non-infant males, Spectrograms: figure 1h, i (see also [16], his fig. 11d).

General structure: simple tonal to complex; 1–3 contiguous units (all howls were measured as a single unit). Frequency modulation: howls typically show a sharp frequency drop after the call onset, followed first by a noisy or low-frequency tonal unit, then by a frequency rise (which usually is not frequency-modulated). Harmonics: 5–7 harmonics; the first two are strongest. Additional characteristics: howls are usually emitted in series, from a few to as many as 30 calls (see [31] for an illustration of a howling bout).

Contexts of emission: adolescent males call with as many repetitions as adults, but juveniles emit fewer calls. Females respond to howls primarily with late-high wails but emit early-high wails and moans as well. Females also respond to the howling of males in other groups. Examples of usage: (a) howling occurs most often as a dusk chorus just prior to the lemurs settling down to sleep in the trees. An adult high-ranking male usually begins, and others join in. Calling reaches a climax after 20–30 s with most males participating; the chorus then begins to dissipate; (b) at various times of the day during the breeding season, males sometimes wander away from the group and emit a single howl, then appear to listen for a response. Inferred function: male advertisement call; together with female counter-calling, howls advertise the presence and location of the group [18].

Vocalization: 'Hm'. Synonyms: 'cohesion grunt' [17]. Vocalizers: all except infants less than about 5 weeks of age. Spectrograms: figure 1j, k.

General structure: simple tonal; single unit. Frequency modulation: exhibits variable degrees of quavering. Harmonics: usually no more than a weak second harmonic, rarely a third. Additional characteristics: develops from the infant trill (see Infant Vocalizations).

Contexts of emission: emitted regularly during, and just prior to, slow group relocation. Examples of usage: (a) uttered during all slow group progressions; (b) given with increasing frequency from the time that group members awaken, begin to stir, groom themselves and begin to move; (c) emitted occasionally by individuals shortly after having been aggressively displaced by a conspecific. Inferred function: (a) indicates that slow group relocation is imminent and promotes group cohesion during the same as an 'acoustic beacon'; (b) reflects a caller's desire to maintain conspecific contact.

Vocalization: 'Huh'. Spectrograms: figure 11, m. Vocalizers: emitted most frequently by infants 3 months of age or older, and by male juveniles and adolescents; given only rarely by adults.

General structure: simple tonal; single unit. Frequency modulation: sharp downslope. Harmonics: variable. Additional characteristics: the fundamental frequency is sometimes filtered out of the call (only calls with intact fundamentals were measured).

Contexts of emission: same as hmm but emitted much less frequently. Because the acoustic structure of the huh renders it more localizable than the hmm, emission of huhs by immatures during progressions effectively marks their positions in the moving group. Examples of usage: (a) see examples a and b for hmm; (b) mixed with hmms by Lc1 group young adult and adolescent males while walking along a path next to the intermediary fence (seemingly awaiting the arrival of Lc2 group); each male's pace and direction of travel along the fence was independent of the others, and therefore interindividual distances were frequently greater than during cohesive group progressions; (c) huhs were emitted occasionally during typical group progressions and were given at these times by immatures. Inferred function: similar to hmm but

marks a caller's location more effectively [37].

Vocalization: 'Purr' [18]. Spectrograms: figure 1n. Vocalizers: emitted most frequently by adult females but given by both sexes of all age classes.

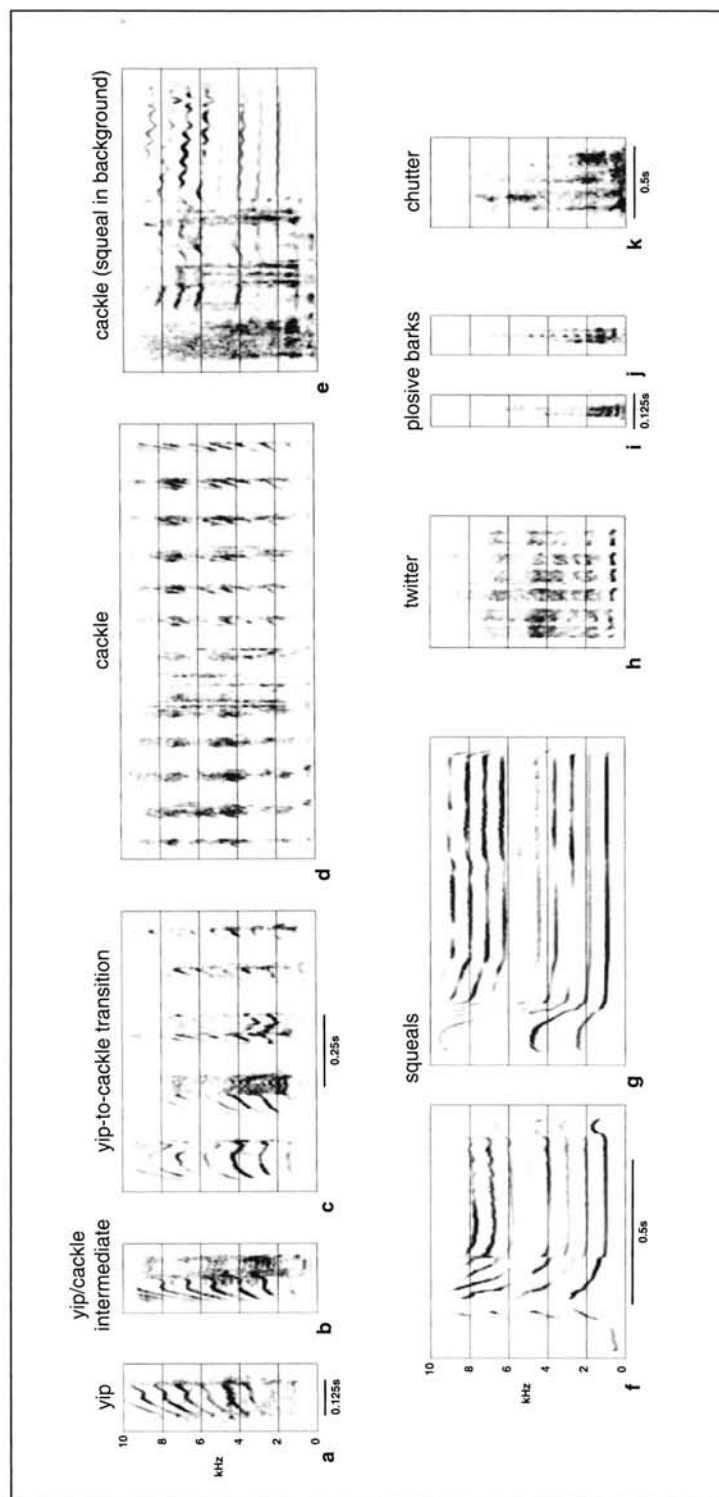
General structure: simple atonal; multi-unit. Spectrographically, purring is seen as a relatively continuous stream of broad-band vertical striations. These striations represent rapid, unvoiced glottal pulses that are emitted while breathing. Additional characteristics: purring sounds like that produced by a domestic cat.

Contexts of emission: most commonly emitted by adults and adolescents when grooming infants (not necessarily their own). Infants purr on occasion when groomed (but more often emit trills). Examples of usage: (a) given by adult females when grooming their own or other infants; (b) emitted occasionally during mutual grooming [18]. Inferred function: appears to express contentment; also may communicate nonaggressive intent of an adult during close contact [M. Pereira, pers. commun.].

Vocalization: 'Chirp'. Spectrograms: figure 1o-r (fig. 1s: chirp/early-high wail intermediate). Vocalizers: all except infants less than about 3 weeks of age.

General structure: simple tonal; single or multi-unit. Frequency modulation: gradual modulations as well as abrupt frequency shifts. Harmonics: most chirps are pure-toned and lack harmonic bands, but a second harmonic is sometimes observed. Additional characteristics: highly variable in structure, chirps are brief tonal elements that often exhibit abrupt frequency shifts. Common forms include (a) single tonal units (fig. 1o, p), (b) pairs or triplets of tonal units (fig. 1r) with each unit being emitted at a different starting frequency and (c) sequences of 5-10 tonal units, each being similar in starting frequency (relatively rare: fig. 1q).

Fig. 2. Agonistic vocalizations.
Legend as in figure 1.



Contexts of emission: emitted during, and just prior to, rapid group movement; chirps elicit early-high wails and moans from conspecifics. Examples of usage: (a) emitted just prior to and during the initial stages of provisioning; (b) given in response to advertisement calls of hawks (e.g. *Buteo* sp.); (c) an individual seemingly motivated to stimulate group relocation will move 20–50 m from the group in the direction of desired travel and emit chirps periodically until the group follows. Inferred function: elicits rapid group movement and may promote group cohesion in this context.

Agonistic Vocalizations

Vocalization: 'Yip' [16]. Synonyms: 'light yip' or spat call [18]. Spectrograms: figure 2a–c; (see also [16]; his fig. 10c, d). Vocalizers: all noninfants, except alpha females.

General structure: simple tonal; single or multi-unit. The yip forms an acoustic continuum with the cackle. Frequency modulation: chevron- or sine-wave-shaped; quavered or unquavered. Harmonics: often 10 or more.

Contexts of emission: emitted singly or in series by subordinates when in close proximity to dominants. Examples of usage: (a) given by a subordinate when approaching, or when approached by, a dominant; (b) most frequently elicited from a subordinate by a cuff (or threat of one) from a dominant. Inferred function: expresses mild fear and, perhaps, willingness to defer to a dominant.

Vocalization: 'Cackle'. Synonyms: 'high-intensity yip' [16]; 'deep spat' [18]. Spectrograms: figure 2d, e. Vocalizers: adults of both sexes.

General structure: simple atonal to complex; multi-unit. Frequency modulation: if visible, FM may be seen as brief upsweeps, downsweeps or chevrons within a cackle unit. Harmonics: numerous, if visible within broad-band frequency distortion.

Contexts of emission: contexts of reluctant submission. Examples of usage: (a) emitted by males while lunging toward tail-waving opponents that are attempting to maintain or reverse rank order. Defensive lunges are sometimes carried to an extreme form in which the challenged male places his open mouth against the side of his tailwaving opponent's face while emitting cackles (fig. 3d); (b) given by subordinate females with newborns when approached by certain dominant females. Subordinate females often perform the full lunging display in this context. Inferred function: a defensive display that may reflect a willingness to become aggressive if pressed.

Vocalization: 'Squeal' [18]. Synonyms: 'sharp call' [16, 17]. Spectrograms: figure 2f, g (see also [17], their fig. 11c). Vocalizers: males, during tail-waving only.

General structure: simple tonal; single unit. Frequency modulation: variable but nearly always an extreme downsweep, or an upsweep followed by a downsweep, at the call's onset. The tonal structure can be quavered or unmodulated. Harmonics: as many as 13 bands can sometimes be seen.

Contexts of emission: occurs only during male tail-waving display, whether asserting status toward a male or toward a female (as part of courtship). Most common in adults but given on occasion by adolescents and juveniles. (When infant males first exhibit tail-waving at about 10 weeks of age, the display is fully adult-like in form but is not accompanied by squeals.) Inferred function: male 'status assertion' vocalization.

Vocalization: 'Twitter' [18]. Spectrograms: figure 2 h. Vocalizers: all except infants less than about 6 months of age.

General structure: simple tonal; multi-unit. The overall structure and the pulsed nature of the twitter are extremely similar to those of the yip. Unlike the yip, however, the terminal frequency of the twitter is



always lower than the initial frequency. Frequency modulation: begins with a slight upsweep in each unit, then follows with a sharp downsweep somewhat like that seen in the huh. Harmonics: usually numerous.

Contexts of emission: assertive contexts in which the caller appears somewhat fearful of reprisal by the target. Emitted infrequently. Examples of usage: (a) used as a vocal 'protest' by mothers not prepared to have their infants

groomed by an individual dominant to them, and (b) as witnessed also by Jolly [18], twitters are sometimes given by males just prior to the squeal when tail-waving. Inferred function: a vocalization that communicates a somewhat fearful but nevertheless assertive demeanor.

Vocalization: 'Plosive Bark'. Spectrograms: figure 2i, j. Vocalizers: both sexes of all age classes.



Fig. 3. **a** Adult male PX (left) squealing and tail-waving at adult male PL. **b** Adult male PL (right) emitting a cackle in response to a squealing/tail-waving display by adult male RO. **c** Adult male DI (right) emitting a cackle in response to a squealing/tail-waving display by adult male HI. **d** DI (right) performing a 'check bite' display toward HI.



General structure: simple atonal; single unit. The plosive bark is a broad-band noise burst with an embedded harmonic structure. Harmonics: 1–5 harmonics appear to be present within the broad-band noise.

Contexts of emission: contexts are few for adults (cf. infants) and involve situations where an intensive vocal threat display is desired. Emission of a plosive bark is usually

accompanied by a lunge toward the eliciting stimulus. Examples of usage: (a) this call is emitted most commonly by young infants, on or off the mother, toward objects (including conspecifics and humans) perceived as threatening. Plosive barks emitted by infants toward conspecifics elicit prompt approach by their mothers, which then may cuff and/or vocally threaten such conspecifics; (b) I wit-

nessed an adult ringtailed lemur emitting plosive barks toward humans only on one occasion, when the individual was being transported to another location in a wire cage; (c) aside from the chirp, the plosive bark was the most common vocalization emitted in response to perched raptors [23]. Inferred function: a high-intensity threat vocalization.

Vocalization: 'Chutter'. Spectrograms: figure 2k. Vocalizers: dominant adults (toward subordinates of all ages).

General structure: simple atonal; multi-unit.

Contexts of emission: low-to-moderate intensity vocal threat always accompanied by a lunge of the dominant individual (the caller) toward a subordinate. Most lunges appeared to be bluffed attacks, or threats, as they were rarely followed by physical aggression once the subordinate had leapt away. Examples of usage: (a) chutters were emitted most commonly by adult females; (b) chutters occurred frequently at provisioning sites, where inter-individual distances were small; (c) males trying to monopolize access to reproductive females gave chutters when chasing competitors away; (d) females and dominant males sometimes emitted chutters in response to squeals directed toward them by subordinates. Inferred function: low-to-moderate threat vocalization. May encourage subordinates to give way to dominants, thereby reaffirming dyadic dominance relationships.

Alerting and Antipredator Vocalizations

Vocalization: 'Gulp'. Spectrograms: figure 4a, b. Vocalizers: all except infants less than about 14 weeks of age.

General structure: simple tonal; single unit. Frequency modulation: in the form of a rounded or partial chevron. Harmonics: most of the energy is concentrated in the fundamental and second harmonic, but up to 5 bands can sometimes be seen.

Contexts of emission: in response to any startling stimulus, whether visual or acoustic [21–23]. Examples of usage: (a) given upon perception of carnivores, raptors (or their calls), rapidly moving humans, any other unusual and/or potentially threatening object or sound, or sudden disturbance of any kind; (b) even the most subtly issued gulps stimulate infants to begin searching actively for their mothers. Inferred function: generalized 'group alert' vocalization.

Vocalization: 'Rasp'. Spectrograms: figure 4c–f. Vocalizers: all except infants.

General structure: simple atonal; multi-unit. Additional characteristics: rasp duration ranges from brief single (fig. 4c) and multiple emissions (fig. 4d) to somewhat longer emissions (fig. 4e). Rasps can intergrade with shrieks (fig. 4f).

Contexts of emission: emitted upon perception of large aerial stimuli (usually raptors). Examples of usage: (a) given by adults in response to airborne birds that are hawk-sized or larger [19, 20]; (b) emitted by adults also in response to rare, large aircraft (low-flying helicopters) but not to common aircraft (jets, low-flying airplanes or high-flying helicopters). Immatures are less discriminating than adults regarding the range of aerial stimuli toward which they emit rasps [19]. Inferred function: aerial predator alarm call.

Vocalization: 'Shriek' [16], variant 1 (Single Frequency Sweep). Synonyms: 'oua-oua' [17]. Spectrograms: figure 4g–j. Vocalizers: all except infants.

General structure: complex; single unit. Frequency modulation: the tonal constituent of the shriek typically rises sharply at the onset, then descends slowly or with a frequency dip toward the end of the call (fig. 4g, h, j). Harmonics: the fundamental contains most of the energy, but a strong second harmonic and 1 or 2 additional weaker harmonics are sometimes seen.

Contexts of emission: emitted, with rare exceptions, only in response to low-flying, large birds. Shrieks occasionally are given in near-perfect synchrony by several callers (fig. 4i). Inferred function: may serve to inform a raptor that it has been seen, and/or may discourage pursuit by intimidation, as well as to broadcast widely that a low-flying raptor has been detected.

Vocalization: 'Shriek' [16], variant 2 (Multiple Frequency Sweeps). Synonyms: 'oua-oua' [17]. Spectrograms: figure 4k, l. Vocalizers: all except infants.

General structure: simple tonal to complex; single unit. Frequency modulation: sine-wave-like but peaks are sharper than troughs. Harmonics: like variant 1, most of the energy is concentrated in the fundamental and second harmonic. Additional characteristics: the two shriek variants appear to form endpoints of an acoustic continuum, although intermediates are less frequently observed.

Contexts of emission: I have recorded this variant only in cases where *L. catta* appeared to perceive that they were under raptorial attack [23]. Inferred function: same as variant 1, except that variant 2 may express the severer or more urgent nature of the aerial predator encounter.

Vocalization: 'Click' [16]. Synonyms: 'click-grunt' [18]. Spectrograms: figure 4m. Vocalizers: all except infants less than 2 weeks of age.

General structure: simple atonal; single unit (see also [16], his fig 10a, b). Clicks consist of a brief, broadband frequency spike and a segment of low-frequency sound energy.

Contexts of emission: clicks are independent sound units that are often repeated at relatively regular intervals in response to a low-level mammalian disturbance. Stationary individuals exhibit a 'curious-but-wary' appearance when emitting clicks. Also given during

arboreal locomotion. Examples of usage: (a) emitted upon sighting carnivores; (b) given when approached closely by humans; (c) uttered when moving through the trees; (d) used by mothers to summon infants at play several meters away: the mother's clicking approach causes the infant to discontinue its activity and jump onto her back. Inferred function: the click is a low-arousal 'location marker' that draws attention to a caller. When broadcasting the location of a slowly moving individual in the trees, clicks permit moving foliage, caused by movements of obscured conspecifics, to be distinguished from foliage movement caused by other stimuli.

Vocalization: 'Closed-Mouth Click Series'. Synonyms: 'click series' [18]. Spectrograms: figure 4m (CMCS). Vocalizers: all except infants less than about 2 months of age.

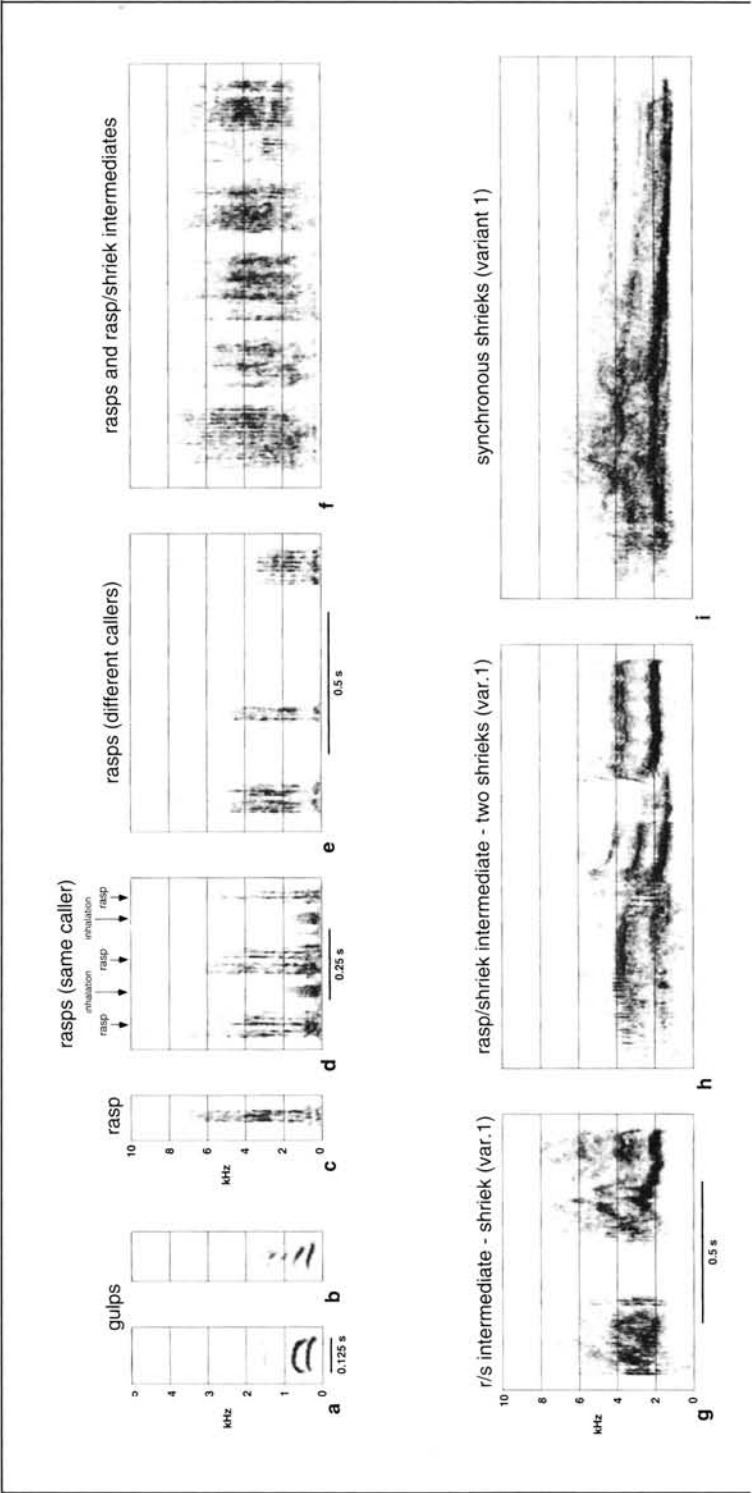
General structure: simple atonal; multi-unit. The CMCS is a pulse train of brief frequency spikes that are similar to clicks but lack the concentration of low-frequency energy observed in the former.

Contexts of emission: given during rapid locomotion (particularly arboreal) and as part of the sequence of calls emitted in response to carnivores. Examples of usage: (a) given when leaping in the trees and/or when landing from a leap; (b) uttered when changing locations in the trees during mobbing of a carnivore. Inferred function: a moderate-arousal 'location marker'.

Vocalization: 'Open-Mouth Click Series'. Synonyms: 'click-series' [18]. Spectrograms: figure 4m (OMCS). Vocalizers: all except infants.

General structure: simple atonal; multi-unit. Additional characteristics: structurally similar to the CMCS, except that concentration of energy is shifted upward by opening the mouth.

Contexts of emission: given during very high levels of arousal. Most frequent in contexts involving carnivores or other mammals.



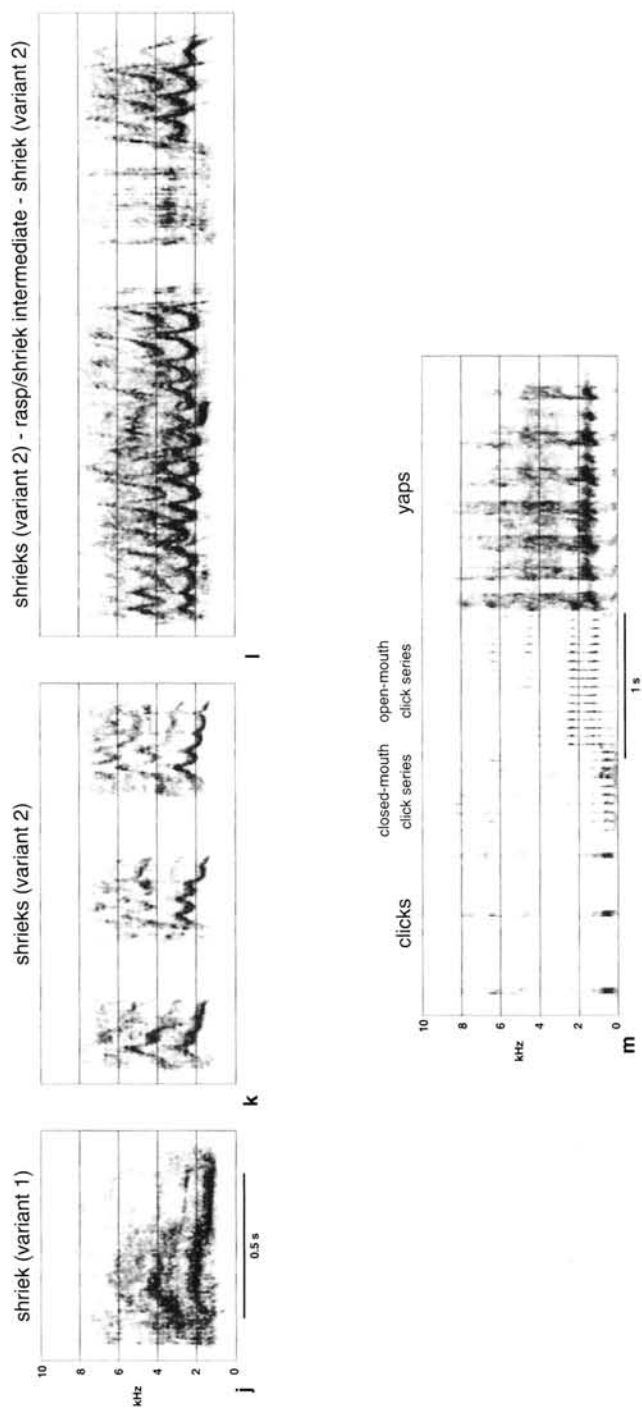


Fig. 4. Alerting and antipredator vocalizations. Legend as in figure 1.

Examples of usage: (a) observed as a structural link between the CMCS and yaps when mobbing carnivores; (b) given on occasion by some members of Lc1 and Lc2 groups when meeting at the fence separating them; (c) emitted by mothers during frenetic chases when trying to retrieve their infant from the back of an unwilling (and rapidly moving) transport. Inferred function: a 'location marker' reserved for a limited number of contexts of very high arousal; the OMCS also appears to serve as a temporal cue that aids the synchronization of yaps.

Vocalization: 'Yap' [18]. Synonyms: 'bark' [16, 17]. Spectrograms: figure 4m. Vocalizers: all but infants.

General structure: complex; single, repeated unit. Tonal chevron embedded in broad-band noise. Frequency modulation: roughly chevron-shaped. Harmonics: all but fundamental are often obliterated by broad-band frequency distortion. If harmonics are visible, up to 7 may be seen.

Contexts of emission: given as a 'mobbing call' toward mammalian stimuli perceived as potentially threatening. Examples of usage: individuals gather around (usually above) the stimulus, while emitting clicks, CMCS and OMCS. Periodic bursts of yaps are issued toward the stimulus in relative synchrony while callers frequently change their positions in the trees. Inferred function: carnivore mobbing call.

Infant Vocalizations

Affiliative Vocalizations

Vocalization: 'Infant Contact Call'. Spectrograms: figure 5a-d.

Ontogenetic period represented in statistical sample (Appendix): day 1 to week 4. General structure: simple tonal; single unit. Frequency modulation: highly variable. Harmonics: largely absent at birth, increas-

ing in number with age. Additional characteristics: this vocalization is the developmental precursor of several adult calls (moan, wails and chirp). Its form is quite variable, although a relatively short duration and a fundamental frequency between 1 and 3 kHz are consistent features.

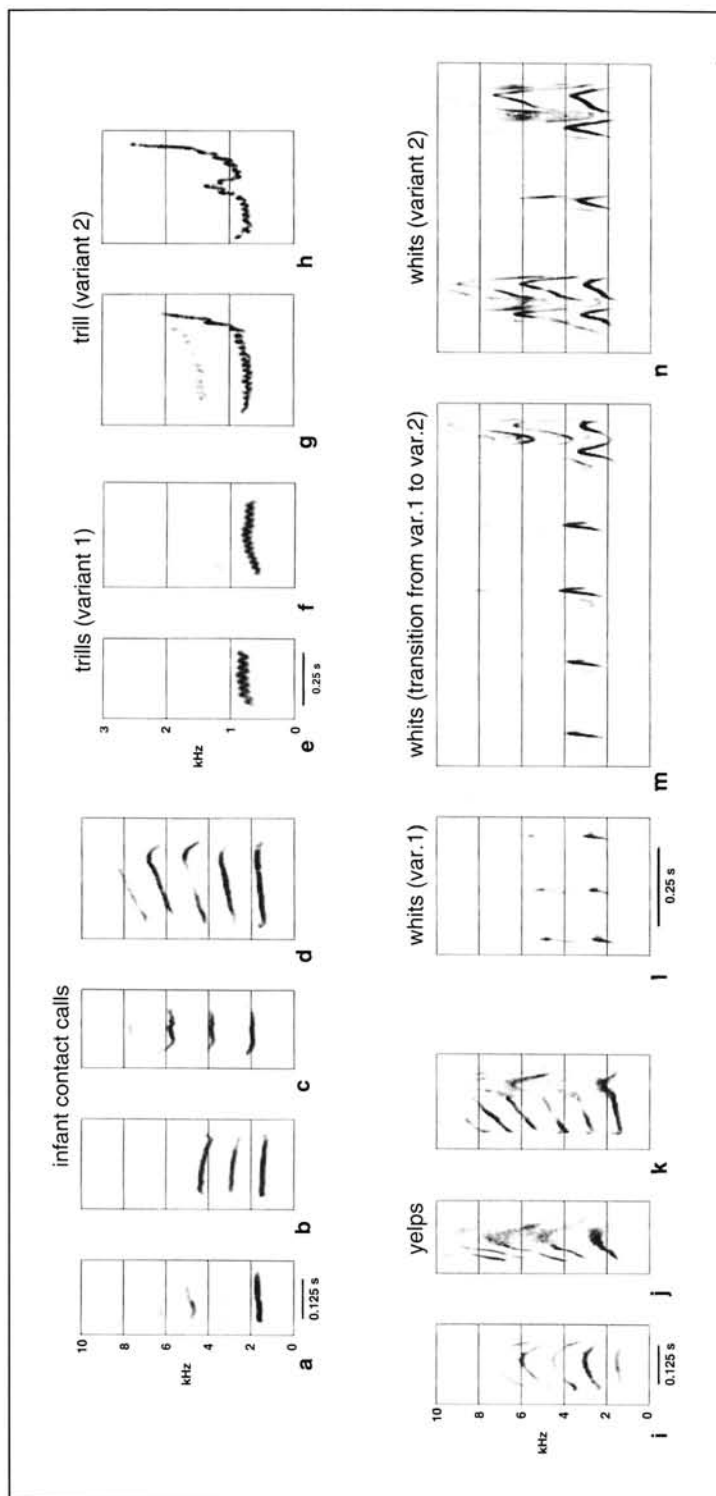
Contexts of emission: initially emitted when an infant attempts to locate and attract the attention of its mother (e.g. when lost or being groomed too roughly by a conspecific). Although this vocalization is later replaced in the repertoire by other call forms, it continues to be issued for some time (at least through the fourth month of life) after the calls that derive from it are being emitted. Examples of usage: (a) given when visual contact with the mother has been lost; (b) eventually emitted antiphonally in response to other affiliative calls. Inferred function: conspecific vocal contact; functions initially to attract the mother and later as a precursor to moans and wails.

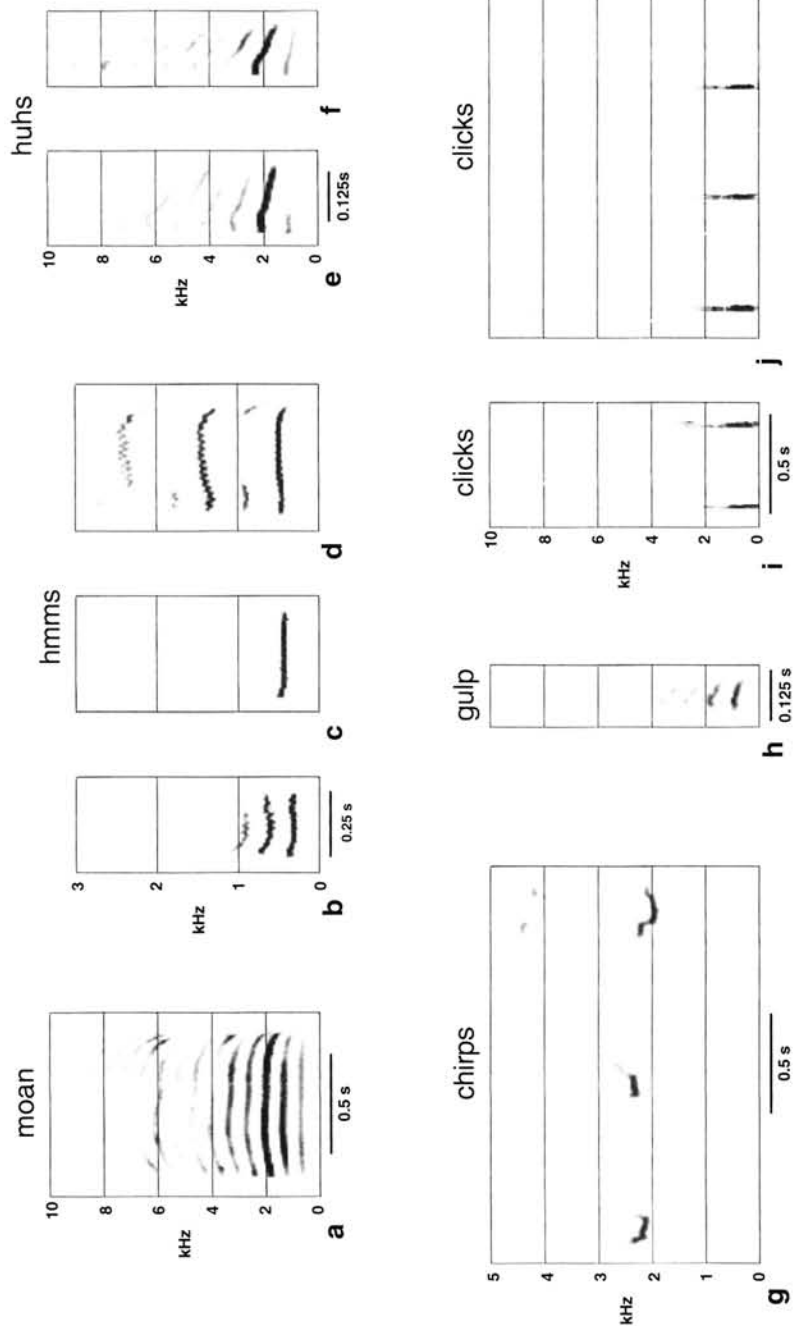
Vocalization: 'Trill', Variant 1. Spectrograms: figure 5e, f.

Ontogenetic period represented in statistical sample (Appendix): day 1 to week 6. General structure: simple tonal, single unit. Frequency modulation: roughly in the form of a sine wave. Harmonics: the fundamental contains most of the call's energy, but a second harmonic is sometimes observed. Additional characteristics: the trill is the developmental precursor of the hmm.

Contexts of emission: emitted by infants during, or in anticipation of, physical contact with conspecifics (particularly the mother). Examples of usage: (a) emitted when auditory or tactile contact with the mother is regained after separation; most commonly given by the infant at the moment it leaps onto its mother's back; (b) infants approximately 2 months of age or older occasionally emit trills during play (juveniles and adolescents emit hmms in the same context); (c) infants continue to issue trills for some time after they are capable of

Fig. 5. Infant vocalizations. Infant contact calls from 3-week-old male DA (**a**, **b**) and 6-week-old male AG (**c**, **d**); trills (variant 1) from 2-week-old male ML or female PY (**e**) and 6-week-old male AG (**f**); trills (variant 2) from female NN at 4 and 5 weeks of age, respectively (**g**, **h**); yelps from 6-month-old male DA (**i**) and 6-week-old male AN (**j**, **k**); whits (variant 1) from 3-week-old male ML or female PY (**l**); whits (transition from variant 1 to variant 2; **m**) and whits (variant 2; **n**) from 1-week-old female SE. Legend as in figure 1.





producing hmms; (d) juveniles often revert to emitting trill-like hmms (fig. 6d) when restrained for examination. Inferred function: expresses desire for, and contentment from, conspecific contact.

Vocalization: 'Trill', Variant 2. Spectrograms: figure 5g, h.

Ontogenetic period represented in statistical sample (Appendix): day 1 to week 6. General structure: similar to variant 1 except that the call terminates with a dramatic rise in frequency. Frequency modulation: roughly sine-wave-shaped, with a strong upward modulation at the offset. Frequency stability is less than in variant 1. Harmonics: 1 or 2 low-energy harmonics are usually seen above the fundamental.

Context of emission: emitted primarily when being groomed anogenitally. Inferred function: may express contentment and/or crossing the sensory threshold from contentment to discomfort.

Distress Vocalizations

Vocalization: 'Whit', Variant 1. Synonyms: 'infant squeaks' [18]. Spectrograms: figure 5l, m.

Ontogenetic period represented in statistical sample (Appendix): day 1 to week 6. General structure: simple tonal; single or multi-unit. Typically seen as a nearly vertical frequency upswing with the major energy being concentrated in the upper portion of the call. Additional characteristics: infant distress calls seem to be conservative in prosimian evolution, as seen in a virtual replica of a whit from a 2-week-old *Galago demidovii* infant ([38], his fig. 68b). Frequency modulation: sharp sweeps across 2–3 kHz. Harmonics: 2 or 3 weak harmonics above the fundamental are observed.

Fig. 6. Infant renditions of adult vocalizations. **a** Moan from 14-week-old male AJ. **b, d** Hmms with trill-like quavering from 15-month-old male ML when restrained for examination. **c** Hmms from 12-month-old male HP. **e, f** Huhs from infant male AJ. **g** Chirps from 7-week-old male BU. **h** Gulp from 15-week-old male AJ. **i** Clicks from 8-week-old male BR. **j** Clicks from 11-week-old female NN. Legend as in figure 1.

Contexts of emission: from general discomfort to distress. Examples of usage: (a) emitted by infants in series in response to uncomfortable physical stimulation, as when being groomed or handled roughly; (b) when unable to reach, or when pushed away from, the nipple. Inferred function: infant distress call; expresses discomfort and/or distress.

Vocalization: 'Whit', Variant 2. Spectrograms: figure 5m (last part) and 5n.

General structure, frequency modulation and harmonics: same as variant 1.

Contexts of emission: variant 2 was observed only in cases of extreme distress. Examples of usage: most commonly emitted when captured and handled by humans. Inferred function: infant high-intensity distress call. Note: structural attributes of variant 2 whits were not quantified because – unlike variant 1, which is relatively uniform in structure – variant 2 whits comprise a series of FM sweeps that are highly variable in 'completeness' and in number. In addition, variant 2 whits suitable for analysis were available from one infant only.)

Vocalization: 'Yelp'. Synonyms: 'high-pitched, modulated plaintive call' [17]. Spectrograms: figure 5i–k. The yelp does not seem to be present at birth and was first clearly identified during the fourth week of life. However, a 3-day-old infant was observed once to emit a yelp-like vocalization.

Ontogenetic period represented in statistical sample (Appendix): weeks 4–16. General structure: simple tonal (sometimes exhibits frequency distortion); single unit. Frequency modulation: roughly chevron-shaped. Harmonics: as many as 8 are frequently visible.

Contexts of emission: extreme fright. Examples of usage: emitted in response to (a) aggression from an unfamiliar adult, (b) having lost contact with the group, (c) being

Table 3. Interstudy comparison of the *L. catta* vocal repertoire

Jolly [18]	This study
Mew	moan
Me-ow	wail (variants 1 and 2)
Howl	howl
Purr	purr
Light yip	yip
Deep spat	cackle
Squeal	squeal
Twitter	twitter
Explosive voiced grunt	plosive bark
Shriek	shriek (variants 1 and 2)
Clicks	click
Click series	click series (open-mouth and closed-mouth)
Yap	yap
Grunt	? (closed-mouth click series?)
Infant squeaks	whit (variants 1 and 2) infant contact call trill (variants 1 and 2) yelp chirp hmm huh chutter gulp rasp

stranded in a location (e.g. the end of a relatively high tree branch) from where the infant appears unable to extricate itself and (d) when captured by a human (or predator?). Inferred function: serves both as an affiliative and distress vocalization in eliciting prompt retrieval by the mother.

(Spectrograms of infant and juvenile renditions of several vocalizations in the adult repertoire are displayed in fig. 6.)

Discussion

The vocal repertoire of the adult ringtailed lemur contains up to 22 distinct call types, with up to 6 additional kinds of vocalizations produced only by infants. An estimate of vocal repertoire size is a consequence of how finely call structure is resolved, however, as well as the researcher's propensity toward 'lumping' or 'splitting' call variants (e.g. see [39]; his table 7). For example, Daschbach et al. [40] reported that the vocal repertoire of the adult slow loris (*Nycticebus coucang*) contained 4 call types, whereas Zimmermann [7] classified the calls into 8 types. Zimmermann's classification contained more calls partly because she described calls not observed by Daschbach et al. [40] and partly because she resolved call structures more finely. Some experimental investigations have tested the species directly for evidence of lumping or splitting of call types that formed acoustic continua [41–43], but such an approach is impossible for entire vocal repertoires.

Jolly [18] reported a vocal repertoire size of 15 call types for *L. catta*. The discrepancy between her assessment and that made here resembles that noted above for *Nycticebus*: (1) I recognize all of the call types that Jolly [18] described verbally (although some of them have been given different names), except for her 'grunt' (table 3), which I was unable to verify (but which might be the closed-mouth click series); (2) I make finer distinctions among 2 call types (wail variants 1 and 2, and shriek variants 1 and 2); (3) 5 of the 9 calls that I describe which are not part of Jolly's [18] account (huh, hmm, trill, chutter, gulp) are emitted at relatively low amplitudes. Given that Jolly [18] usually observed *L. catta* at a distance using binoculars, these calls may have been inaudible to her.

Comparison to Vocal Repertoires of Other Primate Species

In comparison to the vocalizations of other lemur species [16, 17, pers. observation], the ringtailed lemur's vocal repertoire is large and relatively 'discrete' (sensu Marler [44], cf. 'graded') in structure. Of the 28 total call forms (adults plus infants, all variants), structural intermediates were observed only between 8 pairs of call types (thick arrows in fig. 7). Marler [44] initially proposed that discrete vocal repertoires in primates should be expected in forest-dwelling species where line-of-sight distances are short and ambient noise is high: in such habitats communication often depends heavily on vocalization, and readily distinguishable signals could potentially improve communication. In contrast, species living in open environments were predicted to show more intergradations among call types because the constraints imposed by the forested environment would be relaxed, thus permitting visual communication to supplement vocal communication.

In studying vocal communication in several East African colobines [45, 46] and cercopithecines [47], however, Marler found that habitat alone did not sufficiently explain possession of a graded or discrete vocal repertoire. Although three monkey species in the forests of Uganda exhibited largely discrete vocal repertoires (red-tailed monkeys, *Cercopithecus ascanius*; blue monkeys, *C. mitis*; black-and-white colobus, *Colobus guereza*), red colobus (*Colobus badius*) living in the same habitat possessed a highly graded repertoire. This finding led Marler to redefine his hypothesis in terms of social factors. He predicted that as between-group communication increased in importance, vocal repertoires would become more discrete, whereas a greater emphasis on within-group communication would lead to more graded vocal repertoires.

Regarding the ringtailed lemur, no extant prosimian lives in groups as large as those of this species which are relatively stable in size over time (cf. 'fission-fusion') and in which agonistic interaction is typically brief and decisive [48]. Discrete vocal signals and ritualized agonistic visual signals ('tail-waving' display, 'cheek bite' display) therefore might aid in maintaining large-group stability in *L. catta*. Whether the ringtailed lemur emphasizes between-group communication more than within-group communication, at least as compared to other lemur species, cannot be formally addressed due to the current lack of adequate comparative data. I would speculate, however, that this is probably not the case.

In contrast to the ringtailed lemur, the nocturnal, arboreal lorine prosimians exhibit extremely small, highly graded vocal repertoires, with low structural diversity among call types [7, 38, 40]. Whereas the structural gradation may be primitive (e.g. tree shrews [49], squirrels [50], carnivores [51]), small repertoire size in the lorines could be an adaptation to their cryptic life-styles [7, 38]. Nocturnal primates tend to be secretive because nocturnal predators (e.g. owls, viverrids, felids) hunt primarily by sound, not by sight [52]. The nocturnal galagines are more active and more social than lorines and also have more kinds of calls (e.g. 18 call types for *Galago senegalensis senegalensis* [8]). Nevertheless, galagine vocal repertoires, like those of lorines, are relatively limited in overall structural diversity.

In gross terms, *L. catta* compares favorably in structural diversity and repertoire size with many Old World monkeys [53]. Among New World monkeys, cebids tend to exhibit moderately sized, highly graded repertoires [1, 54–56], whereas most callitrichids (though not *Callimico* [57, 58]) exhibit larger, more diverse repertoires. The tamarins (*Saguinus*

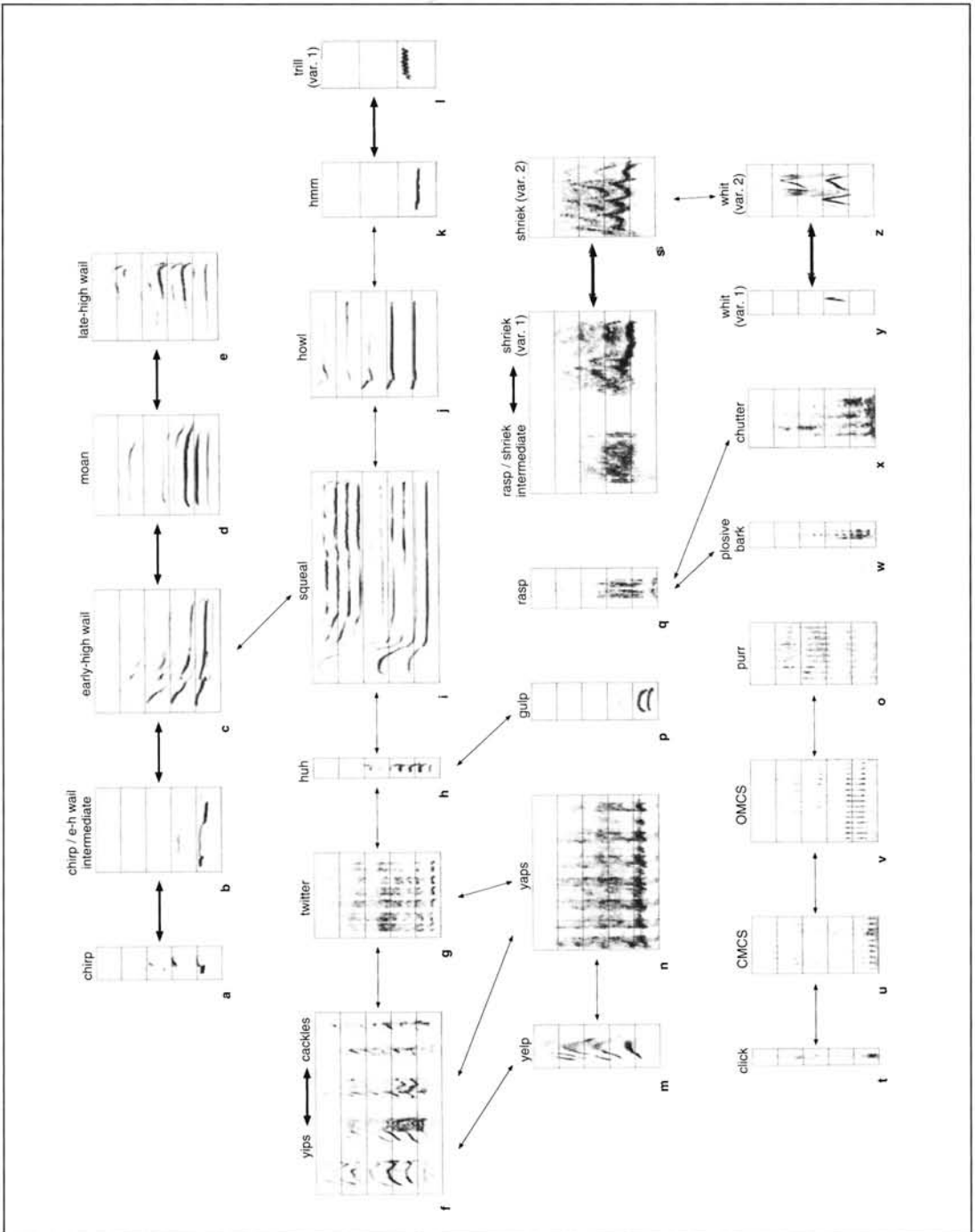


Fig. 7. Diagram of acoustic relationships among the vocalizations of *L. catta*. Thick arrows indicate the existence of structural gradations between call types. Thin arrows indicate some possible relationships between 'discrete' call types that do not intergrade.

sp.), for instance, tend toward very large and structurally diverse vocal repertoires with gradations between some call types but not others [59]. How variation in social parameters influences vocal repertoire size, structural diversity and discrete/graded composition remains a worthwhile topic for future research on nonhuman primate vocal communication.

Preliminary Assessment of Vocal Development in Lemur catta

Although many adult call forms appear in the vocal repertoire by 4 months of age, a thorough documentation of vocal development in this species would require sampling at least through the first year of life. Thus, the data presented here are of limited scope. Some landmarks in *L. catta* vocal development during the first 4 months of life are presented in table 4. The vocalizations produced by immature *L. catta* can be placed into 7 ontogenetic categories such that several trends in the developmental schedules of these calls can be seen (table 5).

Aggressive/defensive calls, whether present at birth or not, are relatively adult-like in structure when they appear in the repertoire. No antipredator calls (i.e. rasp, shriek, yap) were heard during the first 4 months of life (or at any point during the first year). As warning of and defense against predators is the provenance of adults, infant ringtailed lemurs evidently have no need to produce antipredator calls. Neither the huh nor the click are present at birth. Because individuals are usually moving when emitting these calls, the context for their expression (i.e. independent locomotion) is absent during the first few weeks of life. Why these calls should undergo structural modification after they do emerge, however, is not obvious. The howl does not seem to appear until the second year of life and occurs

Table 4. Earliest recording dates for some vocalizations of *L. catta* infants

Week	Vocalization
1 ¹	infant contact call, trills, whits, plosive bark, (purr? yelp?) – see text
3	click (typically appears in week 5)
4	chirp, yelp
5	hmm-like trills (variant 1)
8	moan-like and wail-like infant contact calls
13	huh
14	moan (fully adult form)
15	gulp

¹ Operationally defined as being present at birth.

Table 5. Possible categories of development in *L. catta* vocal repertoire

Calls present at birth

- (1) ... and appear fully adult-like in form: plosive bark (purr?)
- (2) ... but undergo structural change to become *different* calls in adults: infant contact call to moan, wails, and chirp (?); trill (variant 1) to hmm
- (3) ... but may later disappear from the repertoire: whit (variant 1; variant 2?), trill (variant 2), yelp?
- (4) ... and disappear later, then reappear in a modified form and in a different context: whit (variant 2) to shriek (variant 2)?

Calls absent at birth

- (5) ... but are largely adult-like in form when they appear: yip (?), cackle, squeal, twitter, chatter, gulp, CMCS, OMCS, rasp, shriek (both variants) and yap
- (6) ... but undergo some structural modification before reaching adult form: huh, click, howl (yelp to yip?)
- (7) ... and after appearing in the repertoire, later disappear again (yelp? see text)

only in males. The emergence of this advertisement call is likely correlated with pubertal hormonal changes. The vocalization that undergoes the most structural modification in the developing infant is the infant contact call, although it is unclear why one highly variable affiliative vocalization appears to be canalized over time into 4 different types of adult calls (moan, wail variants 1 and 2, chirp).

In general, it seems that calls do not enter the vocal repertoire of *L. catta* until there is some function for them, and other calls seem to vanish when their usefulness is outlived. It is unknown whether calls truly disappear from the vocal repertoire, or are simply no longer expressed. For example, on one occasion a 5-year-old female (BA) responded with yelps to the yelps of her offspring (AG) that had been captured for examination. The mother and infant were about 2 m apart and were separated by a wire mesh screen. This is the only instance in which I witnessed an individual more than 6 months old emitting yelps. Why this adult responded to her infant's yelps by emitting them herself is puzzling, but it demonstrates that at least some infant calls are not actually lost during maturation. This case seems to suggest that it may be the eliciting context, rather than the call itself, which is 'outgrown'.

Comparison with Vocal Development in Other Primate Species

Few data are available on vocal development in prosimians, although Zimmermann has published studies of the mouse lemur (*Microcebus murinus*) [33], the slow loris [60] and the Senegal bushbaby (*G. senegalensis*) [61]. In general, the bushbaby and mouse lemur more closely resemble the ringtailed lemur than does the slow loris in that (1) the infant

vocal repertoires are much smaller than those of conspecific adults, (2) they contain a number of call types that either are unique to infancy or differ considerably in structure from the adult forms, and (3) some calls – particularly those given in affiliative contexts – are far more variable in structure than those given by adults. It is not yet clear whether this structural variation present in infant calls reflects the need for vocal practice or learning [61] or whether it is a consequence of incomplete anatomical and/or neural development. The infant slow loris, in contrast, possesses most of the adult vocal repertoire at or near birth, and these calls typically differ from those of adults only in features associated with the infant's smaller vocal apparatus (e.g. higher call pitch). Differences in vocal development between the slow loris and these other prosimian species may be related to the fact that the slow loris is a relatively unsocial and cryptic primate which possesses a very small vocal repertoire even as an adult [7, 40].

Studies of vocal development in anthropoids tend either to provide detailed data on one type of infant call [62–65] or to scatter information on infant calls throughout descriptions of adult calls. There are a few exceptions, however, in which the development of several call types is examined. One of these is the work by Snowdon and colleagues on the development of the 'closed-mouth trill' and 'J-call' vocalization in the pygmy marmoset (*Cebuella pygmaea*) [66, 67]. Like some calls of infant ringtailed lemurs, mouse lemurs and Senegal bushbabies that are retained into adulthood, the early closed-mouth trills of the pygmy marmoset differ from the same calls produced by adult conspecifics by being higher in pitch, shorter in duration and more irregular in their frequency envelope. Also similar to the affiliative calls produced by the Senegal bushbaby (see fig. 7 in [61]), pygmy marmosets produce long series of calls

(termed 'babbling': see fig. 8 in [66]) where different and highly variable call structures are interspersed in a manner not observed in adults. In contrast to the closed-mouth trill, the J-call did not emerge before week 14, and it was both less variable and more adult-like in structure than the trill when it did appear [66]. The development of the J-call therefore seems similar to the ringtailed lemur huh, for example, which emerges around the same time period and which exhibits minor structural change with maturation (compare fig. 1, panels l and m, with fig. 6e, f).

The comparative development of several vocalizations has also been examined in the vervet monkey (*Cercopithecus aethiops*) [68, 69]. Very much like ringtailed lemur infants, vervets possess at birth several affiliative calls ('grunts' and 'wrrs'), a distress call ('scream') and a threat call ('dominance threat' [69]). Likewise, vervet alarm calls do not seem to be present at birth and, for the most part, are adult-like in structure when they do emerge [68, 69]. Vervet affiliative calls undergo structural changes with maturation, but different calls follow different developmental pathways. The grunts exhibit a slow canalization into adult forms, somewhat similar to the infant contact call and trill variant 1 of *L. catta*. Interestingly, the 'wrr' calls apparently disappear from the vervet vocal repertoire between the ages of 3–10 months, but the 'lost wrr' reappears at 10 months in a slightly modified form and is restricted to the context of intergroup encounters [69]. The only vocalization of *L. catta* that might be analogous here is the whit variant 2 which was emitted by infants only under conditions of extreme distress, and the shriek variant 2 which was emitted by adults in the context of 'attacking hawk' simulations [fig. 7] [19, 23].

Most other studies of vocal development in anthropoids have been conducted on macaques [64, 70–74; but see 75]. Generally

speaking, vocal development in *Macaca* seems similar to that of *L. catta* in that calls emerge and disappear from the repertoire at different times. In *M. arctoides*, 3 call types were observed at birth and a fourth was added in the second week of life [72]. By the end of the sixth month the vocal repertoire contained 12 of the 17 adult call types. Chevalier-Skolnikoff [72] noted that, whereas most infant calls developed into adult calls, some simply vanished from the repertoire. A similar though more complex pattern is apparent in *M. fuscata*, with calls entering and disappearing from the repertoire frequently during development [70, 71].

This commonly observed phenomenon of call types emerging in one week only to disappear a few weeks later argues against an explanation of age-dependent call/context associations. Rather than being linked to a brief window of functional expression, these ephemeral call forms could instead represent a kind of 'phylogenetic baggage' that persists unless major evolutionary reorganization of vocal development occurs. If so, there may be a role for such vocalizations [61] as characters in studies of primate phylogeny.

Acknowledgments

I thank R.H. Wiley for use of his Uniscan II to gather the acoustic data summarized in the Appendix. I also thank K.E. Glander, Director, and E.L. Simons, Scientific Director of the DUPC, for the opportunity to conduct the research presented here. Comments from A. Jolly, M. Hauser, R.D. Martin and several anonymous reviewers improved earlier versions of this manuscript. This research was supported by an NSF Dissertation Improvement Award (BNS 8912589) to J.M.M. This is DUPC publication No. 595.

Appendix: Summary Statistics for the Vocal Repertoire of *L. catta*

IA. Adults: Tonal Calls

Vocalization	Amplitude index	Sample size	Number of individuals sampled			Number of calls per individual		Call duration, s	
			F	M	total	median	range	median	range
Moan	4	124	8	9	17	5	1-36	0.775	0.381-1.525
Wail (1)	5	110	9	8	17	5	1-18	0.666	0.212-1.600
Wail (2)	7-8	12	2	1	3	2	1-9	0.581	0.287-0.731
Hmm	1	46	4	5	9	3	1-15	0.250	0.181-0.412
Huh	2	10	0	4	4	4	1-4	0.112	0.062-0.150
Chirp	2-4	86	4	5	9	5	1-33	0.122	0.056-0.268
Howl	7	110	-	9	9	9	1-37	0.762	0.456-1.293
Squeal	2	6	-	3	3	1	1-4	0.809	0.743-0.987
Gulp	3	24	2	3	5	4	1-13	0.134	0.087-0.225
Shriek (1) ¹	8-9	17	1+	1+	>2	?	?	0.456	0.318-0.868
Shriek (2) ¹	8-9	8	?	?	?	?	?	0.286	0.131-0.525

Vocalization	Fundamental frequency, Hz				lowest frequency				highest frequency			
	start frequency		end frequency		median		range		median		range	
Moan	416	304-960	336	272-736	336	256-544	480	336-960	480	336-960	336-960	336-960
Wail (1)	560	320-2,200	352	240-1,720	336	240-1,640	648	432-2,400	648	432-2,400	432-2,400	432-2,400
Wail (2)	440	360-1,232	384	304-944	360	288-560	576	432-1,160	576	432-1,160	432-1,160	432-1,160
Hmm	416	304-512	352	256-464	304	240-960	452	352-608	452	352-608	352-608	352-608
Huh	600	336-880	400	336-640	360	304-640	680	368-928	680	368-928	368-928	368-928
Chirp	1,320	720-2,200	1,392	784-2,360	1,248	720-1,960	1,512	960-2,440	1,512	960-2,440	960-2,440	960-2,440
Howl	1,040	520-1,880	880	720-960	800	320-960	1,160	920-1,880	1,160	920-1,880	920-1,880	920-1,880
Squeal	400	320-2,240	1,200	880-1,920	320	240-1,200	2,880	2,480-3,440	2,880	2,480-3,440	2,480-3,440	2,480-3,440
Gulp	328	256-528	248	200-352	240	192-336	408	272-640	408	272-640	272-640	272-640
Shriek (1) ¹	1,600	520-5,440	1,360	920-4,160	1,240	520-3,840	2,040	1,160-8,320	2,040	1,160-8,320	1,160-8,320	1,160-8,320
Shriek (2) ¹	2,204	1,540-2,720	1,400	1,040-2,080	1,380	1,040-2,080	3,680	3,040-4,560	3,680	3,040-4,560	3,040-4,560	3,040-4,560

¹ Tonal band only.

IIA. Infants: Infant Tonal Calls

Vocalization	Amplitude index	Number of calls analyzed	Number of individuals sampled			Number of calls per individual		Call duration	
			F	M	total	median	range	median	range
Infant contact call	3-4	146	4	7	11	8	1-42	0.218	0.137-0.450
Trill (1)	2	101	3	5	8	12	1-30	0.256	0.131-0.462
Trill (2)	2	7	2	0	2	-	2-5	0.425	0.312-0.468
Yelp	6	12	1	3	4	3	1-7	0.190	0.125-0.287
Fundamental frequency, Hz									
Vocalization	start frequency		end frequency		lowest frequency		highest frequency		
	median	range	median	range	median	range	median	range	
Infant contact call	2,040	1,080-3,200	1,920	720-3,240	1,800	720-3,040	2,160	1,120-3,760	
Trill (1)	680	400-1,400	760	400-2,200	600	360-1,000	920	512-2,200	
Trill (2)	800	640-960	2,320	1,920-2,680	640	520-840	2,320	1,920-2,680	
Yelp	1,160	840-1,480	1,360	320-2,120	1,120	320-1,480	2,400	1,520-3,840	

IIB. Infants: Infant Atonal and/or Pulsatile Calls

Vocalization	Amplitude index	Number of 'calls' analyzed	Number of individuals sampled				Number of 'calls' per individual	Number of units measured per 'call'		Unit duration, s						
			F	M	un*	total		median	range	n	median	range				
Whit (1)	1-2	29	2	2	1	5	5	1-12	4	1-13	123	0.031	0.012-0.062			
Vocalization	Inter-'call' or interunit interval, s															
	n		median		range		Lowest frequency		Highest frequency		Major energy, Hz		highest frequency			
	n		median		range		median		range		lowest frequency		median		range	
Whit (1)	77	0.137	0.043-0.193		1,200	320-1,960		1,520	360-3,400		2,240	1,240-4,160		1,520	360-3,400	
un. = Sex unknown.																

un. = Sex unknown.

IIIA. Statistics for Infants: Adult Tonal Calls

Vocalization	Number of calls analyzed	Number of individuals sampled			Number of calls per individual		Call duration, s		
		F	M	un.	total	median	range	median	range
Huh	5	0	3	0	3	2	1-2	0.068	0.056-0.100
Chirp	73	3	4	1	8	4	1-25	0.125	0.031-0.350
Vocalization	Fundamental frequency, Hz								
	start frequency		end frequency		lowest frequency		highest frequency		
	median	range	median	range	median	range	median	range	
Huh	800	760-800	520	760-800	520	320-680	840	800-840	
Chirp	1,960	1,088-6,400	1,840	1,072-5,040	1,720	976-4,640	2,120	1,120-6,480	

Age range for recordings: huh, 13 weeks; chirp, 4-8 weeks. un. = Sex unknown.

IIIB. Statistics for Infants: Adult Atonal and/or Pulsatile Calls

Vocalization	Number of 'calls' analyzed	Number of individuals sampled		Number of 'calls' per individual		Unit duration, s			
		F	M	total	median	range	n	median	range
Plosive bark	16	3	6	9	5	1-5	16	0.062	0.050-0.075
Gulp	1	0	1	1	-	-	1	0.112	n/a
Click	27	1	2	3	11	4-15	27	0.031	0.018-0.043
Vocalization	Lowest frequency	highest frequency			Major energy, Hz				
		median	range	median	range	lowest frequency		highest frequency	
						median	range	median	range
Plosive bark	320	240-440	4,040	1,880-9,200	340	240-440	2,000	1,360-3,200	
Gulp	432	-	288	-	272	-	496	-	
Click	280	240-440	2,000	1,000-14,700	280	240-440	1,040	760-1,240	
Age range for recordings: plosive bark, 1-8 weeks; gulp, 15 weeks; click, 5-11 weeks. n/a = Not applicable.									

References

- 1 Robinson JG: Syntactic structures in the vocalization of wedge-capped capuchin monkeys, *Cebus olivaceus*. *Behaviour* 1984;90:46-79.
- 2 Hohmann G: Comparative study of vocal communication in two Asian leaf monkeys, *Presbytis johnii* and *Presbytis entellus*. *Folia Primatol* 1989;52:27-57.
- 3 Hohmann G: Vocal communication of wild bonnet macaques (*Macaca radiata*). *Primates* 1989;30:325-345.
- 4 Hohmann G: Comparative analyses of age- and sex-specific patterns of vocal behavior in four species of Old World monkeys. *Folia Primatol* 1991;56:133-156.
- 5 Hohmann GM, Herzog MO: Vocal communication in lion-tailed macaques (*Macaca silenus*). *Folia Primatol* 1985;45:148-178.
- 6 Newman JD: Squirrel monkey communication; in Rosenblum LA, Coe CL (eds): *Handbook of Squirrel Monkey Research*. New York, Plenum Press, 1985, pp 99-125.
- 7 Zimmermann E: Vocalizations and associated behaviours in adult slow loris (*Nycticebus coucang*). *Folia Primatol* 1985;44:52-64.
- 8 Zimmermann E: The vocal repertoire of the adult Senegal bushbaby (*Galago senegalensis senegalensis*). *Behaviour* 1985;94:212-233.
- 9 Kudo H: The study of vocal communication of wild mandrills in Cameroon in relation to their social structure. *Primates* 1987;28:289-308.
- 10 Pereira ME, Seeligson ML, Macedonia JM: The behavioral repertoire of the black-and-white ruffed lemur (*Varecia variegata variegata*). *Folia Primatol* 1988;51:1-32.
- 11 Aich H, Moos-Heilen R, Zimmermann E: Vocalizations of adult gelada baboons (*Theropithecus gelada*): Acoustic structure and behavior context. *Folia Primatol* 1990;55:109-132.
- 12 Palombi R: A preliminary study of vocal communication in wild long-tailed macaques (*Macaca fascicularis*). I. Vocal repertoire and call emission. *Int J Primatol* 1992;13:143-181.
- 13 Harcourt AH, Stewart KJ, Hauser M: Functions of wild gorilla 'close' calls. I. Repertoire, context, and interspecific comparison. *Behaviour* 1993;124:89-122.
- 14 Stanger KF, Macedonia JM: Vocalizations of the aye-aye (*Daubentonia madagascariensis*) in captivity. *Folia Primatol* 1994;62, in press.
- 15 Snowdon CT: Vocal communication; in Mitchell G, Erwin J (eds): *Comparative Primate Biology*. New York, Liss, 1986, vol 2A: Behavior, Conservation, and Ecology, pp 495-530.
- 16 Andrew RJ: The origins and evolution of calls and facial expressions of the primates. *Behaviour* 1963;20:1-109.
- 17 Petter JJ, Charles-Dominique P: Vocal communication in prosimians; in Doyle GA, Martin RD (eds): *The Study of Prosimian Behavior*. New York, Academic Press, 1979, pp 247-305.
- 18 Jolly A: *Lemur Behavior*. Chicago, University of Chicago Press, 1966.
- 19 Macedonia JM, Polak JF: Visual assessment of avian threat in semi-captive ringtailed lemurs (*Lemur catta*). *Behaviour* 1989;111:291-304.
- 20 Sauther ML: Antipredator behavior in troops of free-ranging *Lemur catta* at Beza Mahafaly special reserve, Madagascar. *Int J Primatol* 1989;10:595-606.
- 21 Macedonia JM: What is communicated in the antipredator calls of lemurs: Evidence from antipredator call playbacks to ringtailed and ruffed lemurs. *Ethology* 1990;86:177-190.
- 22 Macedonia JM: Adaptation and phylogenetic constraints in the antipredator behavior of ringtailed and ruffed lemurs; in Kappeler P, Ganzhorn J (eds): *The Ecological Basis of Lemur Social Systems*. New York, Plenum Press, 1993, pp 67-84.
- 23 Pereira ME, Macedonia JM: Response urgency does not determine antipredator call selection by ringtailed lemurs. *Anim Behav* 1991;41:543-544.
- 24 Macedonia JM, Taylor LL: Subspecific divergence in a loud call of the ruffed lemur (*Varecia variegata*). *Am J Primatol* 1985;9:295-304.
- 25 Macedonia JM: Individuality in a contact call of the ringtailed lemur (*Lemur catta*). *Am J Primatol* 1986;11:163-179.
- 26 Pollock JJ: The song of the indris (*Indri indri*; Primates: Lemnouridae): Natural history, form, and function. *Int J Primatol* 1986;7:225-264.
- 27 Cherry JA, Izard MK, Simons EL: Description of ultrasonic vocalizations of the mouse lemur (*Microcebus murinus*) and the fat-tailed dwarf lemur (*Cheirogaleus medius*). *Am J Primatol* 1987;13:181-185.
- 28 Stanger K: A comparative approach to the organization of vocal signals in nocturnal prosimians. *Hum Evol* 1989;4:181-185.
- 29 Thalmann U, Geissmann T, Simona A, Mutschler T: The indris of Anjanaharibe-Sud, Northern Madagascar. *Int J Primatol* 1993;14:357-381.
- 30 Zimmermann E, Lerch C: The complex acoustic design of an advertisement call in male mouse lemurs (*Microcebus murinus*, Prosimii, Primates) and sources of variation. *Ethology* 1993;93:211-224.
- 31 Macedonia JM, Stanger KF: Phylogeny of the Lemnouridae revisited: Evidence from communication signals. *Folia Primatol*, in press.
- 32 Paillette M, Petter JJ: Vocal repertoire of *Lemur fulvus albifrons*; in Chivers DJ, Herbert J (eds): *Recent Advances in Primatology*. New York, Academic Press, 1978, vol 1, pp 831-834.
- 33 Zimmermann E: Ontogeny of acoustic communication in prosimian primates; in Ehara A, Kimura T, Takenaka O, Iwamoto M (eds): *Primatology Today*. Amsterdam, Elsevier, 1991, pp 337-340.
- 34 Ganzhorn J: Feeding behavior of *Lemur catta* and *Lemur fulvus*. *Int J Primatol* 1986;7:17-29.
- 35 Macedonia JM, Yount PL: Auditory assessment of avian predator threat in semi-captive ringtailed lemurs (*Lemur catta*). *Primates* 1991;32:169-182.

- 36 Watkins WA: The harmonic interval: Fact or artifact in spectral analysis of pulse trains? in Tavolga WN (ed): *Marine Bioacoustics*. New York, Pergamon Press, 1967, vol 2, pp 15–42.
- 37 Snowdon CT, Hodun A: Acoustic adaptations in pygmy marmoset contact calls: Location cues vary with distances between conspecifics. *Behav Ecol Sociobiol* 1981;9:295–300.
- 38 Charles-Dominique P: *Ecology and Behaviour of Nocturnal Primates*. New York, Columbia University Press, 1977.
- 39 Struhsaker TT: Auditory communication among vervet monkeys (*Cercopithecus aethiops*); in Altmann SA (ed): *Social Communication Among Primates*. Chicago, University of Chicago Press, 1967, pp 281–324.
- 40 Daschbach NJ, Schein MW, Haines DE: Vocalizations of the slow loris, *Nycticebus coucang* (Primates, Lorisidae). *Int J Primatol* 1981;2:71–80.
- 41 Snowdon CT, Poly Y: Interspecific and intraspecific responses to synthesized pygmy marmoset vocalizations. *Anim Behav* 1978;26:192–206.
- 42 Mauers M, Streit KM, Barclay D, Wiesner E, Kuchlmorgen B: Interrelations between structure and function in the vocal repertoire of *Saimiri*: Asking the monkeys themselves where to split and where to lump. *Eur Arch Psychiatr Neurol Sci* 1986;236:35–39.
- 43 May B, Moody DB, Stebbins WC: The significant features of Japanese macaque coo sounds: A psychophysical study. *Anim Behav* 1988;36:1432–1444.
- 44 Marler P: Communication in monkeys and apes; in Devore I (ed): *Primate Behavior: Field Studies of Monkeys and Apes*. New York, Holt, Rinehart & Winston, 1965, pp 544–584.
- 45 Marler P: Vocalizations of East African monkeys. I. Red colobus. *Folia Primatol* 1970;13:81–91.
- 46 Marler P: Vocalizations of East African monkeys. II. Black and white colobus. *Behaviour* 1972;42:175–197.
- 47 Marler P: A comparison of vocalizations of red-tailed monkeys and blue monkeys, *Cercopithecus ascanius* and *C. mitis*, in Uganda. *Z Tierpsychol* 1973;33:223–247.
- 48 Pereira ME, Kaufmann R, Kappeler P, Overdorff D: Female dominance does not characterize all of the Lemnidae. *Folia Primatol* 1990;55:96–103.
- 49 Binz H, Zimmermann E: The vocal repertoire of adult tree shrews. *Behaviour* 1989;109:142–162.
- 50 Emmons LH: Sound communication among African rainforest squirrels. *Z Tierpsychol* 1978;47:1–49.
- 51 Peters G: On the structure of friendly close range vocalizations in terrestrial carnivores (Mammalia: Carnivora: Fissipedia). *Z Säugetierkd* 1984;49:157–182.
- 52 Terborgh J, Janson CH: The socioecology of primate groups. *Annu Rev Ecol Syst* 1986;17:111–135.
- 53 Gautier JP, Gautier A: Communication in Old World monkeys; in Sebeok TA (ed): *How Animals Communicate*. Bloomington, Indiana University Press, 1978, pp 890–964.
- 54 Schott D: Quantitative analysis of the vocal repertoire of squirrel monkeys (*Saimiri sciureus*). *Z Tierpsychol* 1975;38:225–250.
- 55 Baldwin JD, Baldwin JJ: Vocalizations of howler monkeys (*Alouatta palliata*) in southwestern Panama. *Folia Primatol* 1976;26:81–108.
- 56 Moynihan M: Communication in the titi monkey, *Callicebus*. *J Zool Lond* 1966;150:77–127.
- 57 Eppe G: Comparative studies on vocalization in marmoset monkeys. *Folia Primatol* 1968;8:1–40.
- 58 Masataka N: A field study on the vocalizations of Goeldi's monkeys (*Callimico goeldii*). *Primates* 1982;23:206–219.
- 59 Cleveland J, Snowdon CT: The complex vocal repertoire of the adult cotton-top tamarin (*Saguinus oedipus*). *Z Tierpsychol* 1982;58:231–270.
- 60 Zimmermann E: Reproduction and behavioral development in the slow loris. *Hum Evol* 1989;4:171–179.
- 61 Zimmermann E: Aspects of reproduction and behavioral and vocal development in Senegal bushbabies (*Galago senegalensis*). *Int J Primatol* 1989;10:1–16.
- 62 Inoue M: Age gradations in vocalization and body weight in Japanese monkeys (*Macaca fuscata*). *Folia Primatol* 1988;51:76–86.
- 63 Lieblich AK, Symmes D, Newman JD, Shapiro M: Development of the isolation peep in laboratory-bred squirrel monkeys. *Anim Behav* 1980;28:1–9.
- 64 Masataka N, Fujita K: Vocal learning of Japanese and rhesus monkeys. *Behaviour* 1989;109:92–199.
- 65 Roush RS, Snowdon CT: Ontogeny of food-associated calls in cotton-top tamarins. *Anim Behav* 1994;47:251–262.
- 66 Snowdon CT: A comparative approach to vocal communication; in Leger DW (ed): *Comparative Perspectives in Modern Psychology*. Nebraska Symposium on Motivation. Lincoln, University of Nebraska Press, 1988, pp 145–199.
- 67 Elowson AM, Snowdon CT, Sweet CJ: Ontogeny of trill and J-call vocalizations in the pygmy marmoset, *Cebuella pygmaea*. *Anim Behav* 1992;43:703–715.
- 68 Seyfarth RM, Cheney DL: Vocal development in vervet monkeys. *Anim Behav* 1986;34:1640–1658.
- 69 Hauser MD: Ontogenetic changes in the comprehension and production of vervet monkey (*Cercopithecus aethiops*) vocalizations. *J Comp Psychol* 1989;103:149–158.
- 70 Kawabe S: Development of vocalization and behavior of Japanese macaques; in Carpenter CR (ed): *Behavioral Regulators of Behavior*. Cranbury, Associated University Press, 1973, pp 164–184.
- 71 Nishimura A: Age-changes of the vocalization in free-ranging Japanese monkeys; in Menzel E (ed): *Symposia of the Fourth International Congress of Primatology*. Basel, Karger, 1973, pp 76–87.
- 72 Chevalier-Skolnikoff S: The Ontogeny of Communication in the Stumptail Macaque (*Macaca arctoides*). *Contrib Primatol*. New York, Karger, 1974, vol. 2.
- 73 Green S: Sex differences and age gradations in vocalizations of Japanese and lion-tailed macaques (*Macaca fuscata* and *Macaca silenus*). *Am Zool* 1981;21:165–183.
- 74 Gouzoules H, Gouzoules S: Design features and developmental modification of pigtail macaque (*Macaca nemestrina*) agonistic screams. *Anim Behav* 1989;37:383–401.
- 75 Gautier JP: Field and laboratory studies of the vocalizations of talapoin monkeys (*Miopithecus talapoin*). *Behaviour* 1974;51:209–273.